

Windows复现graspnet及问题解决

参考博客：[Graspnet手把手复现流程](#) [graspnet复现-CSDN博客](#)

首先git一下graspnet源码，然后开始配置相关GPU环境。

1.安装nvidia驱动和cuda

1. 我的联想拯救者已经自动配置好了nvidia的显卡驱动，所以不用再自己手动安装驱动了，大家可以用 `nvidia-smi` 查看自己的显卡驱动信息，出现以下类似就是显卡驱动安装成功了：

```
Mon Apr 7 22:57:08 2025
+-----+
| NVIDIA-SMI 566.24              Driver Version: 566.24          CUDA Version: 12.7     |
+-----+-----+-----+-----+-----+-----+
| GPU   Name                Driver-Model  Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan   Temp      Perf          Pwr:Usage/Cap       Memory-Usage | GPU-Util  Compute M. |
|=====+=====+=====+=====+=====+=====+
| 0     NVIDIA GeForce RTX 4060 ... WDDM             00000000:01:00.0 On   |          N/A         | |
| N/A    46C    P8              2W / 80W           997MiB / 8188MiB |      1%    Default  |
|                               |                      | N/A         |
+-----+-----+-----+-----+-----+-----+
+-----+
| Processes:                      |
| GPU   GI    CI          PID    Type    Process name                        GPU Memory |
|      ID    ID              |              | Usage   |
+-----+-----+-----+-----+-----+-----+
| 0     N/A    N/A         4028    C+G    ...crosoft\Edge\Application\msedge.exe       N/A      |
| 0     N/A    N/A         6520    C+G    C:\Windows\System32\NahimicSvc64.exe         N/A      |
| 0     N/A    N/A         6572    C+G    C:\Windows\explorer.exe                     N/A      |
| 0     N/A    N/A        11068    C+G    D:\todesk\ToDesk.exe                         N/A      |
| 0     N/A    N/A        14136    C+G    ...1\extracted\runtime\WeChatAppEx.exe       N/A      |
| 0     N/A    N/A        14140    C+G    ...nt.CBS_cw5n1h2txyewy\SearchHost.exe       N/A      |
| 0     N/A    N/A        15468    C+G    ..._8wekyb3d8bbwe\WindowsTerminal.exe       N/A      |
| 0     N/A    N/A        17136    C+G    ...5n1h2txyewy\ShellExperienceHost.exe       N/A      |
| 0     N/A    N/A        22816    C+G    ...vscodel\Microsoft VS Code\Code.exe       N/A      |
| 0     N/A    N/A         88888    C+G    ...ing\Setup\1B...1\Setup\Setup.exe         N/A      |
+-----+-----+-----+-----+-----+-----+
+-----+
```

2. 安装cuda
从 `nvidia-smi` 显示的信息中

安装pytorch

去pytorch的官网查询对应Windows和cuda11.8的pytorch版本，复制命令（后来发现是因为我开了VPN）：

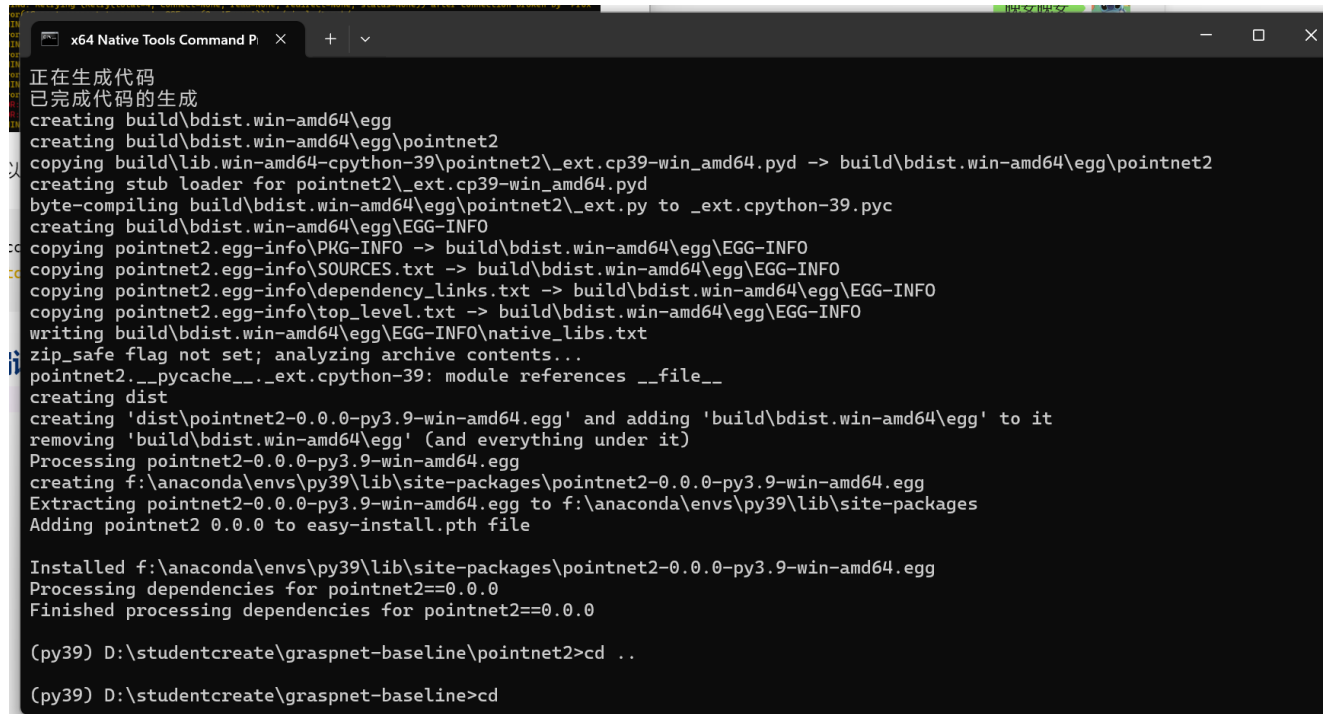
```
WARNING: Retrying (Retry(total=4, connect=None, read=None, redirect=None, status=None)) after connection broken by 'ProxyError('Cannot connect to proxy.', OSError(0, 'Error'))': /simple/torch/
WARNING: Retrying (Retry(total=3, connect=None, read=None, redirect=None, status=None)) after connection broken by 'ProxyError('Cannot connect to proxy.', OSError(0, 'Error'))': /simple/torch/
WARNING: Retrying (Retry(total=2, connect=None, read=None, redirect=None, status=None)) after connection broken by 'ProxyError('Cannot connect to proxy.', OSError(0, 'Error'))': /simple/torch/
WARNING: Retrying (Retry(total=1, connect=None, read=None, redirect=None, status=None)) after connection broken by 'ProxyError('Cannot connect to proxy.', OSError(0, 'Error'))': /simple/torch/
WARNING: Retrying (Retry(total=0, connect=None, read=None, redirect=None, status=None)) after connection broken by 'ProxyError('Cannot connect to proxy.', OSError(0, 'Error'))': /simple/torch/
ERROR: Could not find a version that satisfies the requirement torch==1.13.1 (from versions: none)
ERROR: No matching distribution found for torch==1.13.1
WARNING: There was an error checking the latest version of pip.
```

所以我换成了用conda安装：

SHELL

```
conda install pytorch==1.13.1 torchvision==0.14.1  
torchaudio==0.13.1 pytorch-cuda=11.7 -c pytorch -c nvidia
```

编译pointnet2



```
正在生成代码  
已完成代码的生成  
creating build\bdist.win-amd64\egg  
creating build\bdist.win-amd64\egg\pointnet2  
copying build\lib.win-amd64-cpython-39\pointnet2\_ext.cp39-win_amd64.pyd -> build\bdist.win-amd64\egg\pointnet2  
creating stub loader for pointnet2\_ext.cp39-win_amd64.pyd  
byte-compiling build\bdist.win-amd64\egg\pointnet2\_ext.py to _ext.cpython-39.pyc  
creating build\bdist.win-amd64\egg\EGG-INFO  
copying pointnet2.egg-info\PKG-INFO -> build\bdist.win-amd64\egg\EGG-INFO  
copying pointnet2.egg-info\SOURCES.txt -> build\bdist.win-amd64\egg\EGG-INFO  
copying pointnet2.egg-info\dependency_links.txt -> build\bdist.win-amd64\egg\EGG-INFO  
copying pointnet2.egg-info\top_level.txt -> build\bdist.win-amd64\egg\EGG-INFO  
writing build\bdist.win-amd64\egg\EGG-INFO\native_libs.txt  
zip_safe flag not set; analyzing archive contents...  
pointnet2.__pycache__._ext.cpython-39: module references __file__  
creating dist  
creating 'dist\pointnet2-0.0.0-py3.9-win-amd64.egg' and adding 'build\bdist.win-amd64\egg' to it  
removing 'build\bdist.win-amd64\egg' (and everything under it)  
Processing pointnet2-0.0.0-py3.9-win-amd64.egg  
creating f:\anaconda\envs\py39\lib\site-packages\pointnet2-0.0.0-py3.9-win-amd64.egg  
Extracting pointnet2-0.0.0-py3.9-win-amd64.egg to f:\anaconda\envs\py39\lib\site-packages  
Adding pointnet2 0.0.0 to easy-install.pth file  
  
Installed f:\anaconda\envs\py39\lib\site-packages\pointnet2-0.0.0-py3.9-win-amd64.egg  
Processing dependencies for pointnet2==0.0.0  
Finished processing dependencies for pointnet2==0.0.0  
  
(py39) D:\studentcreate\graspnet-baseline\pointnet2>cd ..  
(py39) D:\studentcreate\graspnet-baseline>cd
```

最终效果如上

在vs的cmd里面执行，然后在conda环境里下载了ninja，原来问题出在ninja，之前编译不成功是因为用的是vs的ninja，后来conda里直接下载ninja是可以自己适配conda里面的cuda和torch的。

1. 打开 **x64 Native Tools Command Prompt for VS 2019** 终端
2. 激活 **py39** 虚拟环境
3. 下载安装ninja

SHELL

```
conda install ninja -c conda-forge
```

4. 检查 **ninja.exe** 的位置

SHELL

```
where ninja.exe # 应输出  
F:\anaconda\envs\py39\Library\bin\ninja.exe
```

5. 强制使用conda的ninja.exe的路径

SHELL

```
set PATH=F:\anaconda\envs\py39\Library\bin;%PATH%
```

6. 设置环境变量为1

SHELL

```
set DISTUTILS_USE_SDK=1
```

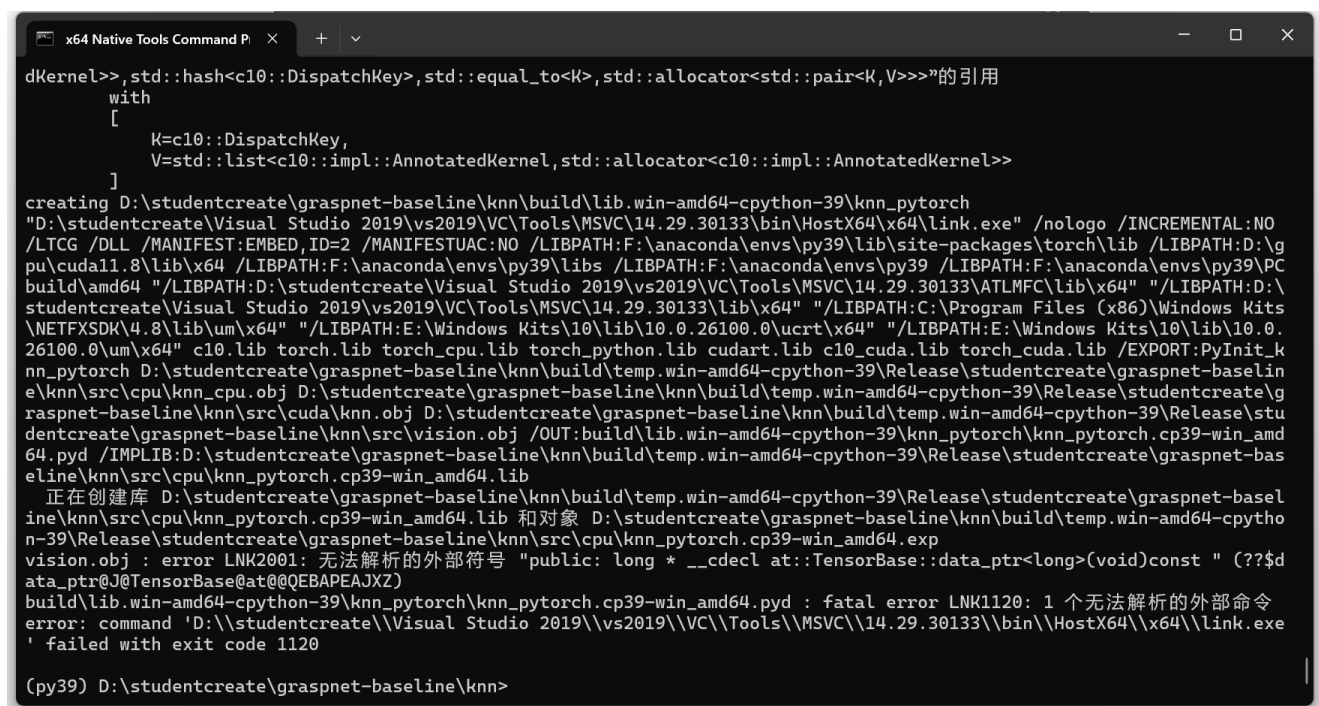
7. 清理和编译

SHELL

```
# 清理旧构建
python setup.py clean --all

# 重新编译 (显示详细日志)
python setup.py install --verbose
```

编译knn



```
dKernel>>,std::hash<c10::DispatchKey>,std::equal_to<K>,std::allocator<std::pair<K,V>>>的引用
with
[
    K=c10::DispatchKey,
    V=std::list<c10::impl::AnnotatedKernel,std::allocator<c10::impl::AnnotatedKernel>>
]
creating D:\studentcreate\graspnet-baseline\knn\build\lib.win-amd64-cpython-39\knn_pytorch
"D:\studentcreate\Visual Studio 2019\vs2019\VC\Tools\MSVC\14.29.30133\bin\HostX64\x64\link.exe" /nologo /INCREMENTAL:NO
/LTCG /DLL /MANIFEST:EMBED,ID=2 /MANIFESTUAC:NO /LIBPATH:F:\anaconda\envs\py39\lib\site-packages\torch\lib /LIBPATH:D:\g
pu\cuda11.8\lib\x64 /LIBPATH:F:\anaconda\envs\py39\libs /LIBPATH:F:\anaconda\envs\py39 /LIBPATH:F:\anaconda\envs\py39\PC
build\amd64 "/LIBPATH:D:\studentcreate\Visual Studio 2019\vs2019\VC\Tools\MSVC\14.29.30133\ATLMFC\lib\x64" "/LIBPATH:D:\
studentcreate\Visual Studio 2019\vs2019\VC\Tools\MSVC\14.29.30133\lib\x64" "/LIBPATH:C:\Program Files (x86)\Windows Kits
\NETFXSDK\4.8\lib\um\x64" "/LIBPATH:E:\Windows Kits\10\lib\10.0.26100.0\ucrt\x64" "/LIBPATH:E:\Windows Kits\10\lib\10.0.
26100.0\um\x64" c10.lib torch.lib torch_cpu.lib torch_python.lib cudart.lib c10_cuda.lib torch_cuda.lib /EXPORT:PyInit_k
nn_pytorch D:\studentcreate\graspnet-baseline\knn\build\temp.win-amd64-cpython-39\Release\studentcreate\graspnet-baselin
e\knn\src\cpu\knn_cpu.obj D:\studentcreate\graspnet-baseline\knn\build\temp.win-amd64-cpython-39\Release\studentcreate\g
raspnet-baseline\knn\src\cuda\knn.obj D:\studentcreate\graspnet-baseline\knn\build\temp.win-amd64-cpython-39\Release\stu
dentcreate\graspnet-baseline\knn\src\vision.obj /OUT:build\lib.win-amd64-cpython-39\knn_pytorch\knn_pytorch.cp39-win_amd
64.pyd /IMPLIB:D:\studentcreate\graspnet-baseline\knn\build\temp.win-amd64-cpython-39\Release\studentcreate\graspnet-bas
eline\knn\src\cpu\knn_pytorch.cp39-win_amd64.lib
正在创建库 D:\studentcreate\graspnet-baseline\knn\build\temp.win-amd64-cpython-39\Release\studentcreate\graspnet-basel
ine\knn\src\cpu\knn_pytorch.cp39-win_amd64.lib 和对象 D:\studentcreate\graspnet-baseline\knn\build\temp.win-amd64-cpytho
n-39\Release\studentcreate\graspnet-baseline\knn\src\cpu\knn_pytorch.cp39-win_amd64.exp
vision.obj : error LNK2001: 无法解析的外部符号 "public: long * __cdecl at::TensorBase::data_ptr<long>(void)const " (??$d
ata_ptr@J@TensorBase@@@QEBAPEAJXZ)
build\lib.win-amd64-cpython-39\knn_pytorch\knn_pytorch.cp39-win_amd64.pyd : fatal error LNK1120: 1 个无法解析的外部命令
error: command 'D:\\studentcreate\\Visual Studio 2019\\vs2019\\VC\\Tools\\MSVC\\14.29.30133\\bin\\HostX64\\x64\\link.exe
' failed with exit code 1120

(py39) D:\studentcreate\graspnet-baseline\knn>
```

然后我把knn文件夹里所有long都改成了int,

(参考博客: [Graspnet在Windows下复现成功-CSDN博客](#))

还是出现以下错误:

```

ninja: build stopped: subcommand failed.
Traceback (most recent call last):
  File "F:\anaconda\envs\py39\lib\site-packages\torch\utils\cpp_extension.py", line 1893, in _run_ninja_build
    subprocess.run(
  File "F:\anaconda\envs\py39\lib\subprocess.py", line 528, in run
    raise CalledProcessError(retcode, process.args,
subprocess.CalledProcessError: Command '['ninja', '-v']' returned non-zero exit status 1.

```

The above exception was the direct cause of the following exception:

```

Traceback (most recent call last):
  File "D:\studentcreate\graspnet-baseline\knn\setup.py", line 58, in <module>
    setup(
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\_init_.py", line 117, in setup
    return distutils.core.setup(**attrs)
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\_distutils\core.py", line 186, in setup
    return run_commands(dist)
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\_distutils\core.py", line 202, in run_commands
    dist.run_commands()
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\_distutils\dist.py", line 983, in run_commands
    self.run_command(cmd)
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\dist.py", line 999, in run_command
    super().run_command(command)
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\_distutils\dist.py", line 1002, in run_command
    cmd_obj.run()
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\command\install.py", line 109, in run
    self.do_egg_install()
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\command\install.py", line 167, in do_egg_install
    self.run_command('bdist_egg')
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\_distutils\cmd.py", line 339, in run_command
    self.distribution.run_command(command)
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\dist.py", line 999, in run_command
    super().run_command(command)
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\_distutils\dist.py", line 1002, in run_command
    cmd_obj.run()
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\command\bdist_egg.py", line 177, in run
    cmd = self.call_command('install_lib', warn_dir=False)
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\command\bdist_egg.py", line 163, in call_command

```

```

    cmd_obj.run()
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\command\bdist_egg.py", line 177, in run
    cmd = self.call_command('install_lib', warn_dir=False)
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\command\bdist_egg.py", line 163, in call_command
    self.run_command(cmdname)
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\_distutils\cmd.py", line 339, in run_command
    self.distribution.run_command(command)
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\dist.py", line 999, in run_command
    super().run_command(command)
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\_distutils\dist.py", line 1002, in run_command
    cmd_obj.run()
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\command\install_lib.py", line 19, in run
    self.build()
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\_distutils\command\install_lib.py", line 110, in build
    self.run_command('build_ext')
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\_distutils\cmd.py", line 339, in run_command
    self.distribution.run_command(command)
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\dist.py", line 999, in run_command
    super().run_command(command)
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\_distutils\dist.py", line 1002, in run_command
    cmd_obj.run()
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\command\build_ext.py", line 99, in run
    _build_ext.run(self)
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\_distutils\command\build_ext.py", line 365, in run
    self.build_extensions()
  File "F:\anaconda\envs\py39\lib\site-packages\torch\utils\cpp_extension.py", line 843, in build_extensions
    build_ext.build_extensions(self)
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\command\build_ext.py", line 481, in build_extensions
    self._build_extensions_serial()
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\command\build_ext.py", line 507, in _build_extensions_serial
    self.build_extension(ext)
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\command\build_ext.py", line 264, in build_extension
    _build_ext.build_extension(self, ext)
  File "F:\anaconda\envs\py39\lib\site-packages\setuptools\command\build_ext.py", line 562, in build_extension
    objects = self.compiler.compile(
  File "F:\anaconda\envs\py39\lib\site-packages\torch\utils\cpp_extension.py", line 815, in win_wrap_ninja_compile
    _write_ninja_file_and_compile_objects(
  File "F:\anaconda\envs\py39\lib\site-packages\torch\utils\cpp_extension.py", line 1574, in _write_ninja_file_and_compile_objects
    _run_ninja_build(
  File "F:\anaconda\envs\py39\lib\site-packages\torch\utils\cpp_extension.py", line 1909, in _run_ninja_build
    raise RuntimeError(message) from e
RuntimeError: Error compiling objects for extension

```

修改如下：

把knn.h中的 **at::** 全改成了 **torch::**，因为这是2.0的pytorch的新规范；
 还改了 **torch::kint** 为 **torch::kInt32**，是PyTorch 2.0中标准的整数类型定义。
 然后才编译成功。

运行demo

```
sh command_demo.sh
```

SHELL

出现如下结果：（一开始正面是没有抓取点位的，用鼠标拖动翻转一下就可以看到。）

